BRAZIL CAR DATA COLLECTION PROJECT

Summer, 2026

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PROJECT OBJECTIVE

The objective of this project is to build a parcel-level environmental dataset for Brazil by combining:

1. Geospatial boundary files from CAR
2. Parcel-level environmental information from the CAR consultation portal

Final output:

Geospatial Parcel Data
+
CAR Environmental Records

Research Dataset

DATA SOURCES

Source 1: Geospatial Downloads

https://consultapublica.car.gov.br/publico/estados/downloads

Source 2: CAR Consultation Portal

https://www.car.gov.br/#/consultar

PHASE 1: UNDERSTAND DATA ARCHITECTURE

For Acre:

Download:

Perímetros dos imóveis

This dataset contains parcel geometries and a unique identifier:

Codigo_IMOVEL

This identifier is the key that links:

Geospatial parcel
↔
CAR consultation record

PHASE 2: CURRENT CODE ARCHITECTURE

The current codebase consists of three major components.

A. Utility Layer

Functions include:

- HTTP requests
- User-agent rotation
- HTML parsing
- Download handling
- Retry mechanisms
- Data extraction helpers

Purpose:

Provide reliable communication with CAR servers.

B. CAR Scraper

The scraper:

1. Accepts CAR identifiers.
2. Solves reCAPTCHA.
3. Requests parcel information.
4. Receives JSON response.
5. Extracts environmental attributes.
6. Writes structured output.

The JSON endpoint returns detailed parcel information.

Important fields include:

- Property status
- Registration status
- Municipality
- State
- Property area
- Native vegetation area
- Legal reserve information
- Protected area information
- Environmental restrictions
- Registration dates

C. Output Pipeline

Outputs include:

- JSON archive files
- CSV files
- Excel exports

This allows:

- Reproducibility
- Validation
- Data auditing

PHASE 3: ACRE PILOT

Step 1

Download Acre parcel boundaries.

Step 2

Load shapefile/geospatial file.

Step 3

Extract Codigo_IMOVEL.

Step 4

Create list of unique parcel IDs.

Step 5

Run scraper against parcel IDs.

Step 6

Store returned JSON.

Step 7

Create cleaned CSV.

Step 8

Perform quality checks.

PHASE 4: SCALE TO ALL STATES

Brazil contains 27 states.

After Acre validation:

Nilesh:
9 states

Vinay:
9 states

Rishikesh:
9 states

Each person should follow the exact same workflow.

DOCUMENTATION REQUIREMENTS

Every task must be documented.

Maintain:

1. Download log
2. Processing log
3. Error log
4. Data quality log
5. Runtime log

Required information:

- Date
- State
- Files processed
- Number of parcels
- Number successfully scraped
- Number failed
- Notes

DATA SECURITY POLICY

This project contains research data and code developed for Indiana University research purposes.

Required Rules:

1. Do not share data externally.
2. Do not share code externally.
3. Do not upload to public GitHub.
4. Do not distribute scripts.
5. Do not share credentials.
6. Do not publish intermediate datasets.
7. Keep all files in approved project storage.

CODE MANAGEMENT RULES

Before modifying code:

1. Create backup.
2. Document change.
3. Test on small sample.
4. Verify outputs.
5. Commit changes with clear notes.

Never modify production scripts without testing.

QUALITY CONTROL

Before marking a state complete:

Verify:

- Parcel counts
- Missing identifiers
- Duplicate identifiers
- Missing JSON records
- Invalid geometries
- Missing environmental fields

All validation results must be documented.

PROJECT SUCCESS CRITERIA

Success means:

1. Acre pipeline fully operational.
2. Reliable extraction of parcel records.
3. Scalable workflow across all states.
4. Complete documentation.
5. Secure handling of all data and code.
6. Reproducible research workflow.

Primary Immediate Goal:

Get Acre working end-to-end as quickly as possible.